Worksheet 20

November 11, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Establish an identity

Prove the identity on its common domain:

$$\frac{\sec x - \cos x}{\tan x} = \sin x.$$

(a) State the restrictions needed for the original expression.

(b) Rewrite the left side using sine and cosine, then simplify to the right side.

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November 11, 2026 Name: _____

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2. Use a conjugate

Prove the identity on its common domain:

$$\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}.$$

(a) State the common-domain restriction.

(b) Multiply the left side by a suitable form of 1 and simplify.